Overview

Convolution
- Simplicial Fourier transform, Frequency, Fourier basis
- Convolutional filters
- Generalizes graph convolutions

Gaussian Process
- How to define GPs for different parts?
- Hodge-compositional idea
- Probabilistic methods

SC ~~ Hodge decomposition

Convolutional NNs
- Architecture (attention, message passing)
- Equivariance, Robustness
- Higher-order link predictions
- Generalizes GCN, Chebnet, etc.

Generative learning
- (Dynamic) optimal transport (Schrödinger bridge) on SCs
- Gaussian bridge
- Generative models: diffusion, flow models on SCs

Some ideas on biomolecular data

- Encode both geometric and (richer) topological information
- Multimodal data, e.g., LLM
- Towards foundation models for drug discovery, e.g., prior-fitted networks

Simplicial (Complex) Convolutional NNs

[Yang et al. 2025 TMLR]

Simplicial Convolutions

[Yang et al. 2022 IEEE TSP]

Topological Schrödinger Bridge Matching

[Yang 2025 ICLR spotlight]